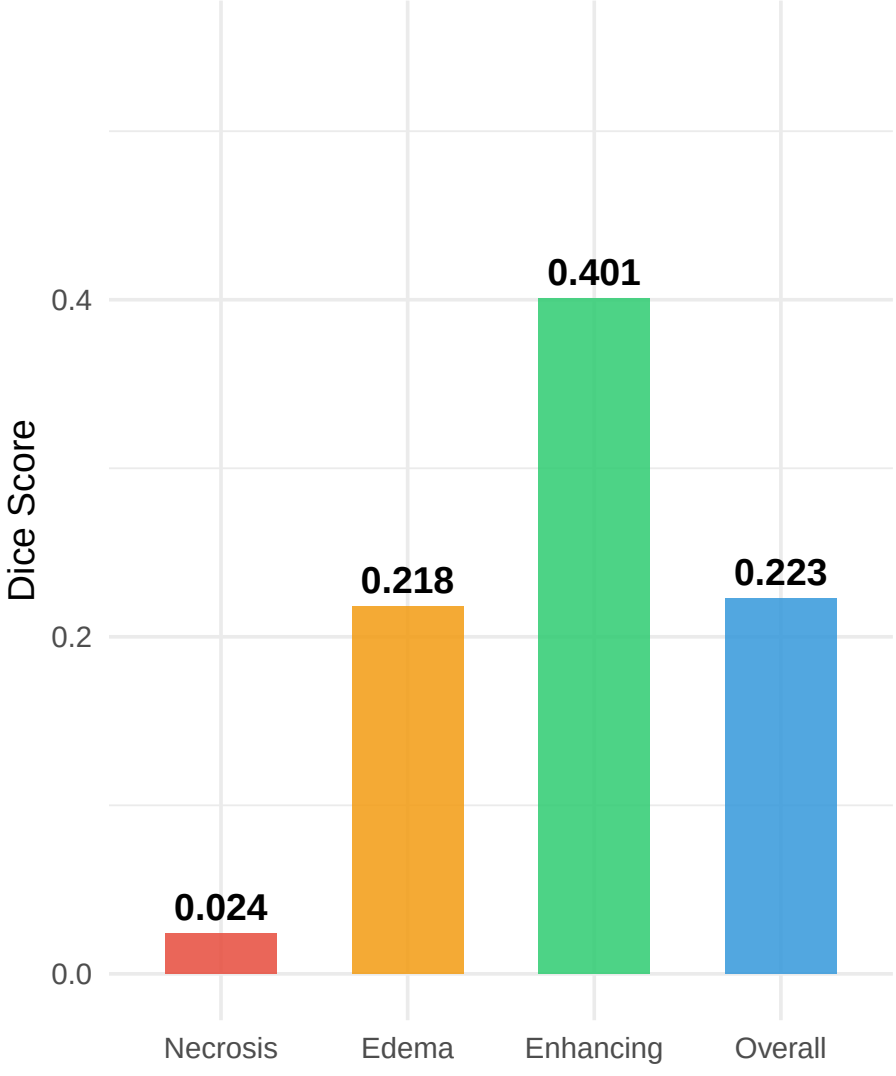
3D U−Net for Tumor Segmentation — LUMIERE Dataset

3D U−Net Segmentation — Per−Class Dice Scores

Trained on 599 samples (91 patients) | 100 epochs | T4 GPU



3D U−Net Architecture Summ

Input: 4−ch MRI (48³)

Output: 4−class seg

Parameters: 5,649,204

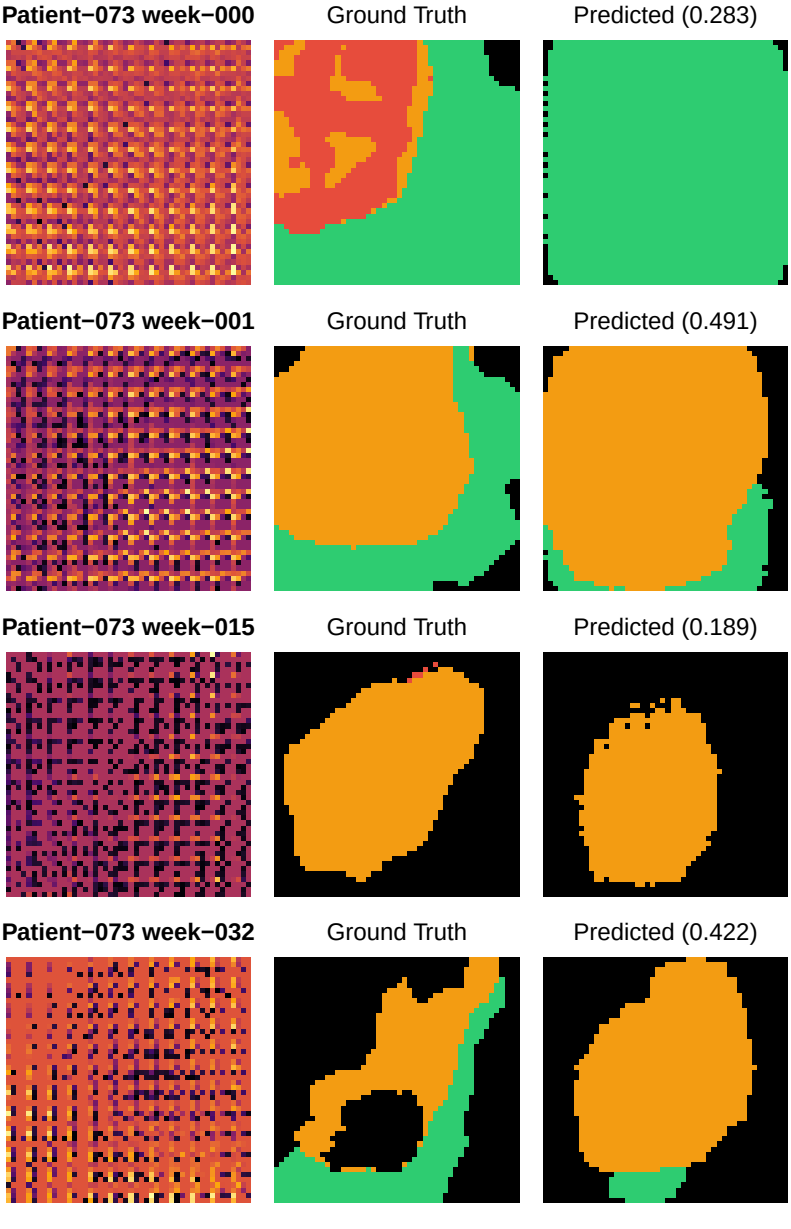
Encoder: 4 levels (16−>128)

Bottleneck: 256 filters

Loss Function: Dice + CE

Optimizer: Adam (lr=1e−4)

Best Val Dice: 0.2733

3D U−Net Inference: FLAIR | Ground Truth | Prediction

Axial slice at z=24 for validation patients